15 GITHUB REPOSITORIES

FOR

AI AGENT DEVELOPERS

SLIDE TO EXPLORE

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12-week curriculum. Classic ML. Project-based. Quiz-backed.

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Machine Learning for Beginners - A Curriculum

> 🌍 Travel around the world as we explore Machine Learning by means of world cultures 🌍

Cloud Advocates at Microsoft are pleased to offer a 12-week, 26-lesson curriculum all about **Machine Learning**. In this curriculum, you will learn about what is sometimes called **classic machine learning**, using primarily Scikit-learn as a library and avoiding deep learning, which is covered in our [AI for Beginners' curriculum]. Pair these lessons with our ['Data Science for Beginners' curriculum], as well!

Travel with us around the world as we apply these classic techniques to data from many areas of the world. Each lesson includes pre- and post-lesson quizzes, written instructions to complete the lesson, a solution, an assignment, and more. Our project-based pedagogy allows you to learn while building, a proven way for new skills to 'stick'.

✍️ **Hearty thanks to our authors** Jen Looper, Stephen Howell, Francesca Lazzeri, Tomomi Imura, Cassie Breviu, Dmitry Soshnikov, Chris Noring, Anirban Mukherjee, Ornella Altunyan, Ruth Yakubu and Amy Boyd

🎨 **Thanks as well to our illustrators** Tomomi Imura, Dasani Madipalli, and Jen Looper

🙏 **Special thanks 🙏 to our Microsoft Student Ambassador authors, reviewers, and content contributors**, notably Rishit Dagli, Muhammad Sakib Khan Inan, Rohan Raj, Alexandru Petrescu, Abhishek Jaiswal, Nawrin Tabassum, Ioan Samuila, and Snigdha Agarwal

🤩 **Extra gratitude to Microsoft Student Ambassadors Eric Wanjau, Jasleen Sondhi, and Vidushi Gupta for our R lessons!**

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A no-fluff ML roadmap built for self-learners.

machine learning for software engineers

Stars 757 | Subscribe | More learning resources | Follow @loganthorneloe | Subscribers 59

Welcome to the **Machine Learning for Software Engineers** repo! ML for SWEs is a newsletter to understand machine learning engineering and gain confidence building with AI. This repo will contain all hands-on newsletter resources for understanding machine learning engineering and building with AI. Currently included are:

- The ML Road Map: Learn ML fundamentals for free from high-quality resources.
- The Machine Learner's Library: Essential books machine learning engineers should read to learn the topics they need to know about building machine learning systems.
- MNIST on MLX: Train a simple model on Apple's machine learning framework designed for Apple silicon.

More to come in the future. I plan to write many more articles with accompanying hands-on activities.

In the meantime, these articles are community favorites:

- Devin Has Exposed a Major Issue with Software Engineering
- Machine Learning Infrastructure: The Bridge Between Software Engineering and AI
- If you want to learn machine learning engineering, start here
- If You Understand Bananas, You Can Understand Machine Learning

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You can find more Society's Backend articles here and you subscribe here to get all articles in your inbox.

Build neural nets from scratch. Understand every step.

Neural Networks: Zero to Hero

A course on neural networks that starts all the way at the basics. The course is a series of YouTube videos where we code and train neural networks together. The Jupyter notebooks we build in the videos are then captured here inside the lectures directory. Every lecture also has a set of exercises included in the video description. (This may grow into something more respectable).

Lecture 1: The spelled-out intro to neural networks and backpropagation: building micrograd

Backpropagation and training of neural networks. Assumes basic knowledge of Python and a vague recollection of calculus from high school.

- YouTube video lecture
- Jupyter notebook files
- micrograd Github repo

Lecture 2: The spelled-out intro to language modeling: building makemore

We implement a bigram character-level language model, which we will further complexify in followup videos into a modern Transformer language model, like GPT. In this video, the focus is on (1) introducing torch.Tensor and its subtleties and use in efficiently evaluating neural networks and (2) the overall framework of language modeling that includes model training, sampling, and the evaluation of a loss (e.g. the negative log likelihood for classification).

- YouTube video lecture
- Jupyter notebook files
- makemore Github repo

The best CV collection: books, datasets, research tips, projects.

Awesome Computer Vision: awesome

A curated list of awesome computer vision resources, inspired by awesome-php.

For a list people in computer vision listed with their academic genealogy, please visit here

Contributing

Please feel free to send me pull requests or email (jbhuang@vt.edu) to add links.

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Awesome Lists

- Awesome Machine Learning
- Awesome Deep Vision
- Awesome Domain Adaptation
- Awesome Object Detection
- Awesome 3D Machine Learning
- Awesome Action Recognition

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From transformers to multilingual NLP tools, plus top courses.

awesome-nlp

A curated list of resources dedicated to Natural Language Processing

Read this in English, Traditional Chinese

Please read the contribution guidelines before contributing. Please add your favourite NLP resource by raising a pull request

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Understand LLMs from the ground up with really well curated notebooks

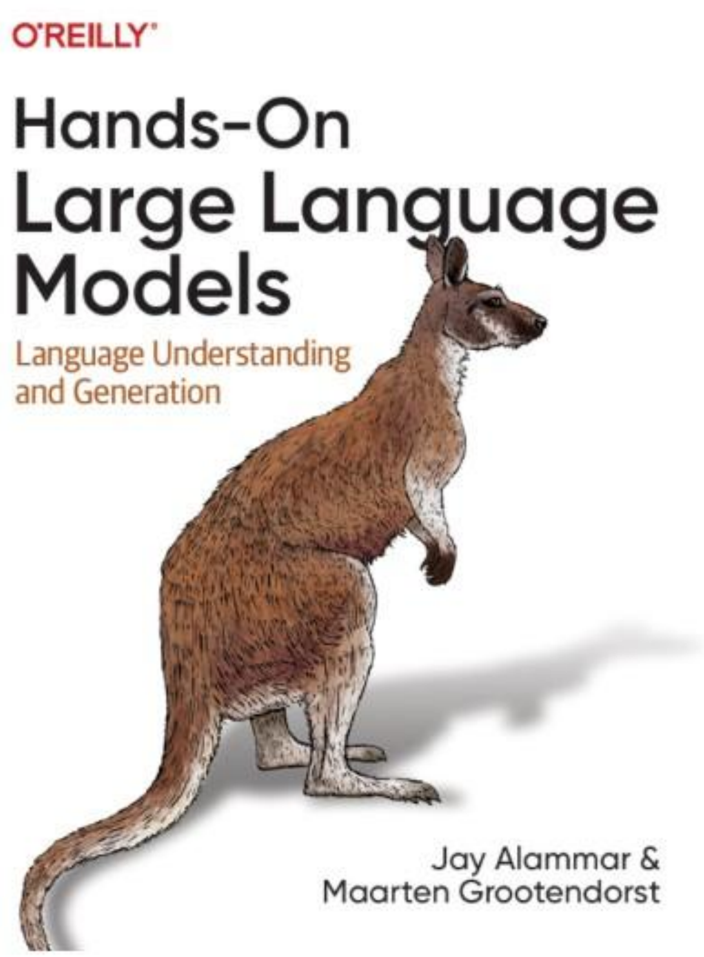
Hands-On Large Language Models

Follow Jay | Follow Maarten | DeepLearning.AI Course NEW!

Welcome! In this repository you will find the code for all examples throughout the book Hands-On Large Language Models written by Jay Alammar and Maarten Grootendorst which we playfully dubbed:

"The Illustrated LLM Book"

Through the visually educational nature of this book and with **almost 300 custom made figures**, learn the practical tools and concepts you need to use Large Language Models today!



The book is available on:

- Amazon
- Shroff Publishers (India)
- O'Reilly
- Kindle
- Barnes and Noble
- Goodreads

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Everything prompt-engineering under the sun in one place.

Prompt Engineering Guide

Prompt engineering is a relatively new discipline for developing and optimizing prompts to efficiently use language models (LMs) for a wide variety of applications and research topics. Prompt engineering skills help to better understand the capabilities and limitations of large language models (LLMs). Researchers use prompt engineering to improve the capacity of LLMs on a wide range of common and complex tasks such as question answering and arithmetic reasoning. Developers use prompt engineering to design robust and effective prompting techniques that interface with LLMs and other tools.

Motivated by the high interest in developing with LLMs, we have created this new prompt engineering guide that contains all the latest papers, learning guides, lectures, references, and tools related to prompt engineering for LLMs.

🌐 Prompt Engineering Guide (Web Version)

🎉 We are excited to launch our brand new prompt engineering courses under the DAIR.AI Academy. Join Now!

Use code PROMPTING20 to get an extra 20% off.

Happy Prompting!

Announcements / Updates

- 🎓 We now offer self-paced prompt engineering courses under our DAIR.AI Academy. Join Now!
- 🎓 New course on Prompt Engineering for LLMs announced! Enroll here!
- 💼 We now offer several services like corporate training, consulting, and talks.
- 🌐 We now support 13 languages! Welcoming more translations.
- 👩‍🎓 We crossed 3 million learners in January 2024!
- 🎉 We have launched a new web version of the guide here
- 🔥 We reached #1 on Hacker News on 21 Feb 2023
- 🎉 The First Prompt Engineering Lecture went live here

Join our Discord

Follow us on Twitter

Subscribe to our YouTube

Subscribe to our Newsletter

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The ultimate free resource list: books, courses, datasets, tools.

AWESOME DATA SCIENCE

An open-source Data Science repository to learn and apply towards solving real world problems.

This is a shortcut path to start studying **Data Science**. Just follow the steps to answer the questions, "What is Data Science and what should I study to learn Data Science?"

Sponsors

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---	Be the first to sponsor! `github@academic.io`

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Clean RL implementations with a cheat sheet.

All RL Algorithms from Scratch ✨

This repository is a collection of Python implementations of various Reinforcement Learning (RL) algorithms. The *primary* goal is **educational**: to get a deep and intuitive understanding of how these algorithms work under the hood. 🧠 Due to the recent explosion in the AI domain especially Large Language Models, and many more applications it is important to understand core reinforcement learning algorithms.

This repository also includes a comprehensive cheat sheet summarizing key concepts and algorithms for quick reference.

This is *not* a performance-optimized library! I prioritize readability and clarity over speed and advanced features. Think of it as your interactive textbook for RL.

Updates:

- **2 April 2025**: Added a comprehensive RL Cheat Sheet summarizing all implemented algorithms and core concepts. Repository now includes 18 algorithm notebooks.
- **30 March 2025**: Added 18 new algorithms.

🌟 Why This Repo?

- **Focus on Fundamentals:** Learn the core logic *without* the abstraction of complex RL libraries. We use basic libraries (NumPy, Matplotlib, PyTorch) to get our hands dirty. 🛠️
- **Beginner-Friendly:** Step-by-step explanations guide you through each algorithm, even if you're new to RL. 👶
- **Interactive Learning:** Jupyter Notebooks provide a playground for experimentation. Tweak hyperparameters, modify the code, and *see* what happens! 🧪
- **Clear and Concise Code:** We strive for readable code that closely mirrors the mathematical descriptions of the algorithms. No unnecessary complexity! 👌
- **Quick Reference:** Includes a detailed Cheat Sheet for fast lookups of formulas, pseudocode, and concepts.

A rich archive of RL theory, papers, demos, and apps.

Awesome Reinforcement Learning awesome

This page is no longer maintained.

A curated list of resources dedicated to reinforcement learning.

We have pages for other topics: awesome-rnn, awesome-deep-vision, awesome-random-forest

Maintainers: Hyunsoo Kim, Jiwon Kim

Contributing

Please feel free to pull requests

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Most current GenAI list: courses, tools, and papers by recency.

Awesome Generative AI awesome Track Awesome List

> A curated list of Generative AI projects, tools, artworks, and models

- Generative AI Area
  - Generative AI history, timelines, maps, and definitions
  - Ethics, Philosophical questions and Discussions about Generative AI
  - Critical Views about Generative AI
  - Generative AI Processes and Artifacts
  - Generative AI Tools Directories
  - Courses and Educational Materials
  - Human-AI Interaction
  - Papers Collection
  - Online Tools and Applications
- Code and Programming
  - Vibe Coding
  - AI-Powered Code Generation
- Text
  - Everything to Markdown to LLMs
  - Small Language Models
  - Large Language Models (LLMs)
    - Model Context Protocol
    - Programming Frameworks for LLMs
    - Prompt Engineering
      - Prompt Optimizers
      - Prompt Engineering for Text-to-text
      - Prompt Engineering for Text-to-image
    - Mamba
    - Running LLMs Locally
    - Function Calling
    - GPTs and Assistant API
    - Retrieval-Augmented Generation (RAG)
    - Embeddings and Semantic Search
    - Autonomous LLM Agents
      - Multi-agents
    - LLM Evaluation
    - LLMOps
    - AI Engineering

Jump into agentic systems with a free course.

AI Agents for Beginners - A Course

11 Lessons teaching everything you need to know to start building AI Agents

license MIT | contributors 42 | issues 8 open | pull requests 2 open | PRs welcome

🌐 Multi-Language Support

Supported via GitHub Action (Automated & Always Up-to-Date)

English | Chinese Simplified | Chinese Traditional
Chinese Hong Kong | French | Japanese | Korean | Portuguese Brazilian | Spanish | German
Persian | Polish | Hindi

Watch 278 | Fork 7.7k | Star 28k

Azure AI Foundry | 27350 MEMBERS

👉 Understand the bones of RAG and become a retrieval master.

Advanced RAG Techniques: Elevating Your Retrieval-Augmented Generation Systems 🚀

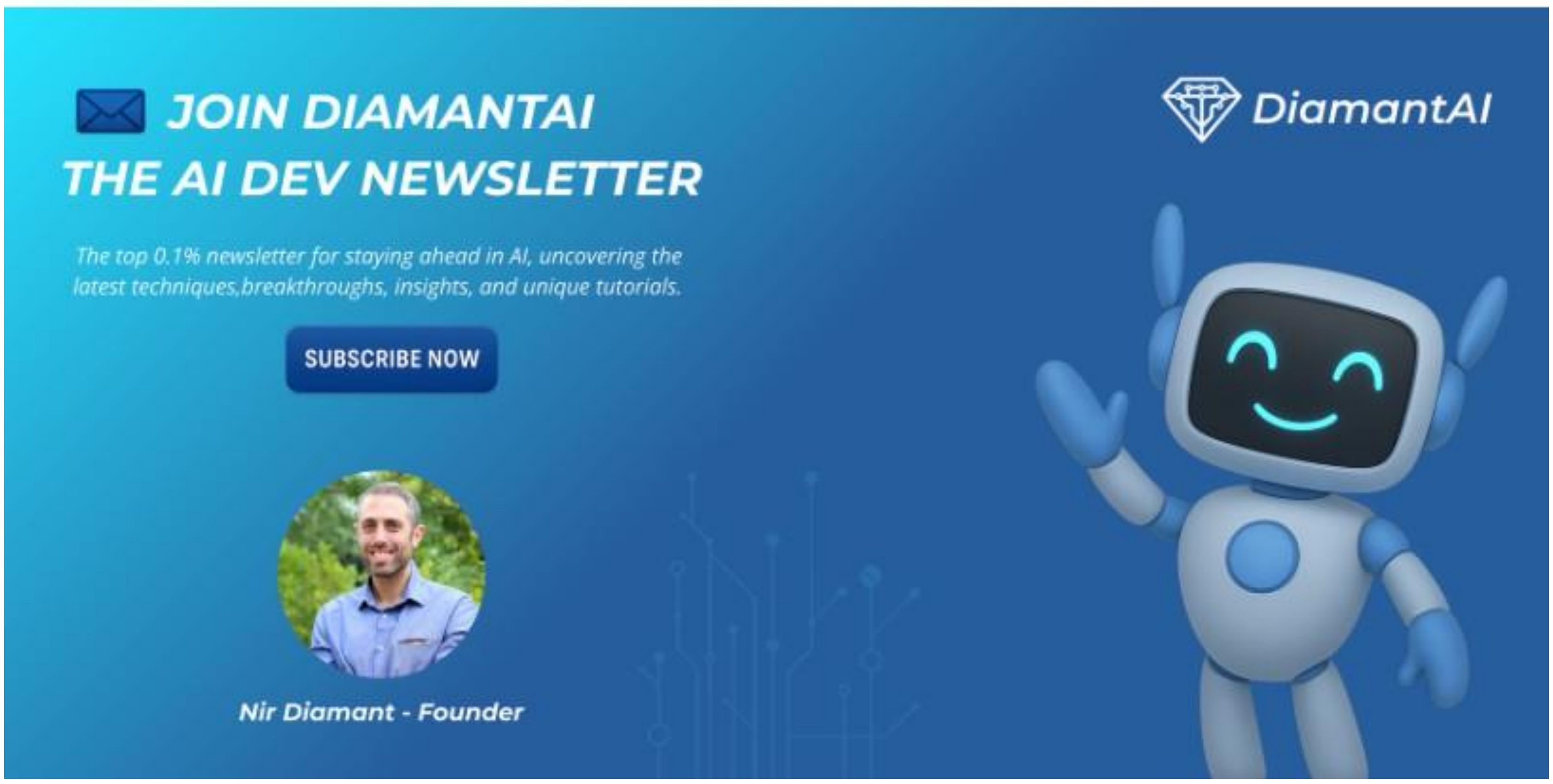
Welcome to one of the most comprehensive and dynamic collections of Retrieval-Augmented Generation (RAG) tutorials available today. This repository serves as a hub for cutting-edge techniques aimed at enhancing the accuracy, efficiency, and contextual richness of RAG systems.

📫 Stay Updated!

🚀 Cutting-edge Updates	💡 Expert Insights	🎯 Top 0.1% Content

Subscribe to Newsletter

Join over 20,000 of AI enthusiasts getting unique cutting-edge insights and free tutorials! ***Plus, subscribers get exclusive early access and special 33% discounts to my book and the upcoming RAG Techniques course!***



GenAI Agents: Comprehensive Repository for Development and Implementation 🚀

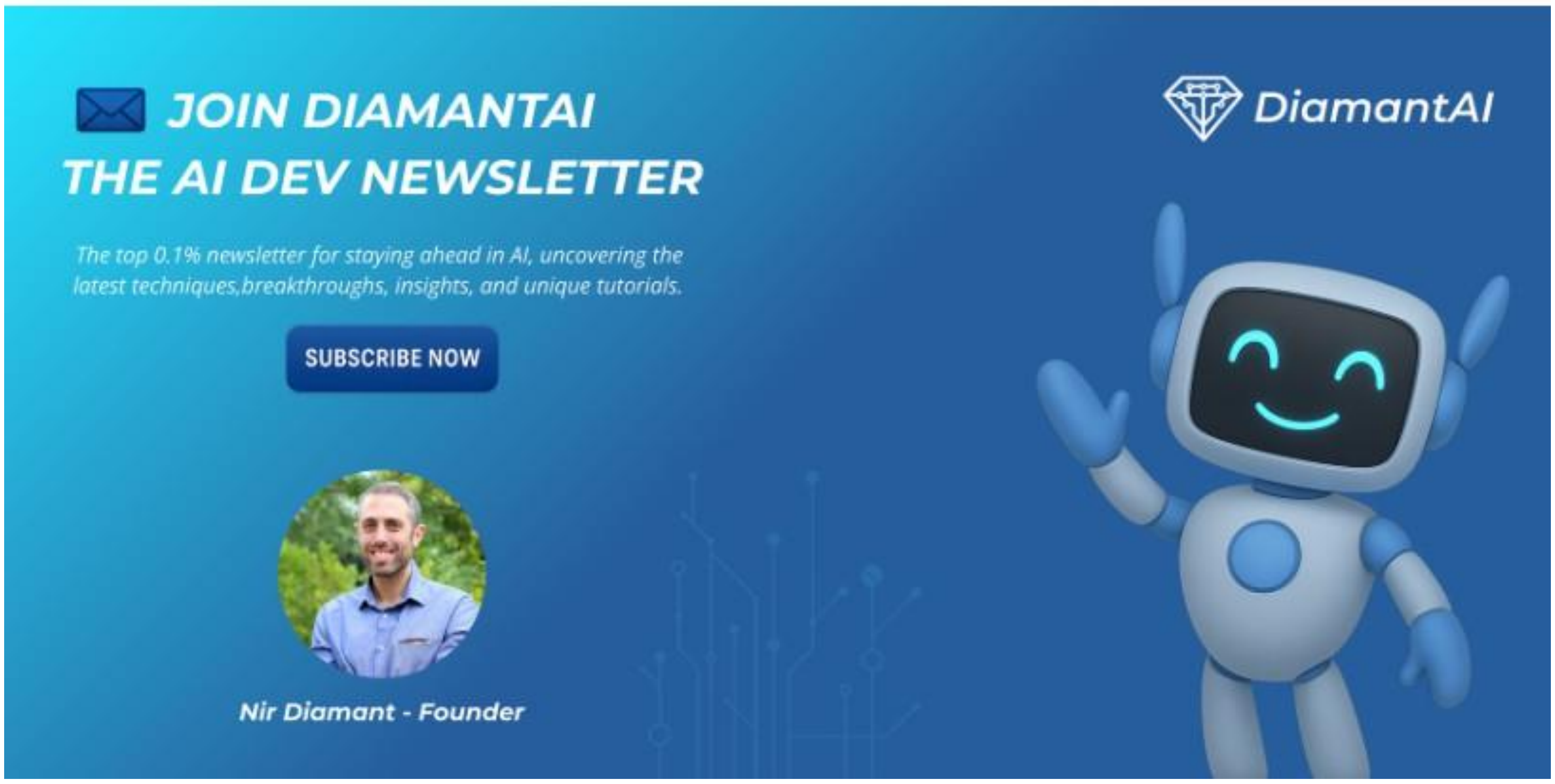
Welcome to one of the most extensive and dynamic collections of Generative AI (GenAI) agent tutorials and implementations available today. This repository serves as a comprehensive resource for learning, building, and sharing GenAI agents, ranging from simple conversational bots to complex, multi-agent systems.

📫 Stay Updated!

🚀 Cutting-edge Updates	💡 Expert Insights	🎯 Top 0.1% Content

Subscribe to Newsletter

Join over 20,000 of AI enthusiasts getting unique cutting-edge insights and free tutorials! ***Plus, subscribers get exclusive early access and special 33% discounts to my book and the upcoming RAG Techniques course!***



The definitive MLOps guide for any practitioner.

Made With ML

Design · Develop · Deploy · Iterate
Join 40K+ developers in learning how to responsibly deliver value with ML.

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🔥 Among the top ML repositories on GitHub

Lessons

Learn how to combine machine learning with software engineering to design, develop, deploy and iterate on production-grade ML applications.

- Lessons: https://madewithml.com/
- Code: GokuMohandas/Made-With-ML

1. 🎨 **Design**
   - Setup
   - Product
   - Systems
2. 🔢 **Data**
   - Preparation
   - Exploration
   - Preprocessing
   - Distributed
3. 🤖 **Model**
   - Training
   - Tracking
   - Tuning
   - Evaluation
   - Serving
4. 💻 **Develop**
   - Scripting
   - Command-line
5. 📦 **Utilities**
   - Logging
   - Documentation
   - Styling
   - Pre-commit
6. 🧪 **Test**
   - Code
   - Data
   - Models
7. ♻️ **Reproducibility**
   - Versioning
8. 🚀 **Production**
   - Jobs & Services
   - CI/CD workflows
   - Monitoring
   - Data engineering

Live cohort

Sign up for our upcoming live cohort, where we'll provide the **structure, compute (GPUs)** and **community** to learn everything in one weekend.

Learn more

IntereSted in
more content like thiS?

FolloW me :

Sachin Patil